Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Full-Stack Web Application Development · Enterprise Systems · Database & APIs · Technical Leadership · 9yr Production

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PROFESSIONAL SUMMARY

Senior software engineer with 9 years designing, developing, deploying, and supporting enterprise web applications and data systems at production scale. At Entrata, I owned 20+ full-stack production modules used by 100K+ monthly active users -- designing solutions for complex problems independently, contributing to all phases of the software development lifecycle, and leading peers through code reviews and mentorship. I build reliable, scalable, and secure applications: multi-tenant PostgreSQL schemas optimized across 200+ queries, encrypted AWS data pipelines with 100% PHI compliance, Docker microservices with 50% scalability improvement, and RabbitMQ async systems designed with idempotency guarantees. I am quick to learn new platforms and architectures -- at Entrata I became one of the deepest technical owners of a large proprietary enterprise SaaS platform, understanding its architecture, APIs, data structures, and business processes from the ground up. Currently completing M.S. Data Analytics Engineering at George Mason University (GPA 3.87, May 2026), where I also work as a Research Associate building data pipelines and dashboards for faculty leadership -- giving me direct experience with the strategic planning, stakeholder environment, and full stack technical needs of an academic institution. Portfolio: venkatalakshmiparimala.github.io/Portfolio/about.html

TECHNICAL SKILLS

Languages & Frameworks: Python (9yr production), JavaScript, TypeScript, SQL/PL-SQL, PHP, R, Bash, HTML/CSS; FastAPI, Flask, React.js, Angular, jQuery; REST APIs, API Integration, GraphQL (familiarity)

Database, Architecture & Systems Design: PostgreSQL (200+ query optimization, stored procedures, multi-tenant schemas, data modeling), MySQL, Oracle, Microsoft SQL Server (T-SQL), NoSQL; Relational Database Design, Data Architecture, ETL/ELT Pipelines, Microservices Architecture, Distributed Systems, Scalable & Extensible Application Design, Performance Engineering, Security & Secure Development, Application/Database Integration, RabbitMQ (async middleware)

Cloud, Infrastructure & Delivery: AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, CI/CD (GitHub Actions), Git, Source Control & Version Control, Agile/Scrum, Sprint Planning, Code Reviews, Testing & Debugging, Change Control, Technical Documentation & Best Practices, Production Support & Troubleshooting; AI/ML: LangGraph, LLM APIs, RAG Pipelines, Agentic Systems, LLM Evaluation

EXPERIENCE

Entrata -- Multifamily Real Estate SaaS (PropTech)

Sep 2018 -- Aug 2024

Senior Software Engineer -- Enterprise Web Applications, Database Systems & Platform Engineering · Lehi, UT (Remote)

- **Designed and developed 20+ full-stack, web-based production modules independently** -- Python, PHP, JavaScript, PostgreSQL -- serving 100K+ monthly active users across a multi-tenant enterprise platform; contributed across all phases of the SDLC (design, development, testing, deployment, production support) for 120+ Agile sprints; maintained 99.5% uptime across compliance-critical systems.
- **Owned complex enterprise systems end-to-end** -- long-term sole owner of the annual HUD income-limit pipeline: designed the solution architecture, wrote the data mapping and transformation logic, defined validation and reconciliation standards, and managed every release with rollback preparation. Thousands of federal source records expanded into ~180,000 domain records powering eligibility, rent, and tax-credit calculations for 100K+ housing units. Zero compliance violations across 7 years of ownership.
- **Database design, architecture, and performance engineering** -- designed and maintained complex multi-tenant relational database schemas for eligibility, financial, and document workflows; optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, partitioning, and stored procedure refactoring; built RabbitMQ async middleware with idempotency guarantees and retry logic eliminating double-processing failures.
- **Led the AWS S3 migration and built secure, reliable infrastructure** -- inventoried all document types, mapped legacy records to new data contracts, validated migrated content, and confirmed all customer-facing workflows post-cutover; Libsodium-encrypted ETL pipelines (100% PHI compliance, zero security incidents); Docker microservices redesign (50% scalability improvement, 30% deployment time reduction).
- **Technical leadership, mentoring junior engineers, and development standards** -- mentored 5+ engineers; conducted 300+ peer code reviews; defined development standards and documentation practices; built internal debugging tools reducing debugging time 40%; kept bug count below 10 for 6+ consecutive months; championed CI/CD and automated testing practices across the team.

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source data pipelines and prediction models (Python, R) across 190+ countries (R^2=0.977); applied Isolation Forest to automate data integrity checks across 3 research pipelines; designed Tableau and ggplot2 dashboards improving research comprehension 40% for faculty leadership; delivered data analytics and programming instruction to 600+ students.

Magizhs Tech Solution

May 2025 -- Aug 2025

Engineer -- AI Systems & Application Development

- Designed and built agentic workflow automation system with multi-step API integration, conditional routing, and validation guardrails; implemented data transformation and quality rules; applied AI automation to improve compliance checking across existing business processes.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Full-Stack Application Development

- Built property management and compliance web application modules (PHP, Python, Java, PostgreSQL); income-limit import pipeline (1.6M records); AWS S3 migration; database schema redesign with post-deployment stabilization across hundreds of thousands of migrated records.

PROJECTS

Leasel: Full-Stack Enterprise Web Application -- End-to-End Ownership

FastAPI · LangGraph · Python · React · PostgreSQL · CI/CD

Python · LangGraph · FastAPI (15-endpoint REST API) · React · SQLite · OCR · GitHub Actions | github.com/venkatalakshmiparimala/Lease_complaince_rag

Designed, developed, tested, and deployed a production web application from scratch: 15-endpoint FastAPI REST API, multi-service orchestration pipeline, CI regression gate, structured logging, and HITL escalation. Demonstrates the full-stack ownership pattern -- design, code, test, deploy, support -- that enterprise portal development requires. 93.6% precision across 1,240 test cases.

AgentProbe: Full-Stack Web Application -- FastAPI Backend & React Frontend

FastAPI · React · TypeScript · CI/CD · PostgreSQL · Automated Testing

Python · FastAPI · React.js · TypeScript · GitHub Actions · PostgreSQL | github.com/venkatalakshmiparimala/AgentProbe

Full-stack web application with FastAPI REST backend, React/TypeScript frontend, CI-integrated regression gate, and structured reporting dashboards. Demonstrates modern web application development practices: API design, frontend development, automated testing, and deployment pipelines -- the same technical stack patterns a modernized student portal requires.

AI on FHIR: Secure Full-Stack Web Platform -- HIPAA & RBAC

Flask · React · REST API · HIPAA · Access Controls · Docker

Python · Flask · React · Docker · FHIR API | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant full-stack web platform: Flask REST API backend, React frontend, RBAC access controls, Docker containerized deployment, REST API integration with external systems. Demonstrates secure application development and the compliance-aware design standards institutional portals require.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017